Same Performance, Different Judgments: Gendered Evaluation Across Success and Failure

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Abstract

Women can be judged more harshly than men for comparable failures, but no two real failures are identical. Using motion-capture animation, we showed the same professional tennis points performed by the original players, avatars of their own gender, a robot, or a single male or female avatar. We asked whether avatars remove any extra credit or blame for women's plays or whether it follows the displayed gender. In two preregistered experiments (2024), U.S. adults sampled with quotas for sex, age, and ethnicity ($N = 3{,}729$; 66% White) rated eight winners for impressiveness or eight errors for severity. Women's errors were rated more severe than men's in the original footage; every avatar version, including same-gender avatars, cut this gap by more than half. If participants perceived the avatars' gender (not measured), displaying it was thus not enough to reproduce how errors were judged in the original footage. Women's winners trailed men's by less in the original footage than with avatars, a less consistent effect, though not significantly smaller than for errors. With only eight points per experiment, and avatar effects nonsignificant with points as a random factor, the conclusions are limited to these points and lay U.S. viewers.

Keywords: gender bias, performance evaluation, double standards, shifting standards, synthetic media

Public Significance Statement

In two experiments, U.S. adults rated video clips of men's and women's professional tennis points, shown either in the original broadcast footage or with the players replaced by animated figures that reproduced their movements. In the original footage, women's mistakes were judged more harshly than men's; with every animated version, even one of the same gender as the real player, more than half of this difference disappeared, so this part appears to stem from how the players were presented rather than from the plays themselves. For settings that show synthetic versions of real people, such as sports media, these findings suggest that whether audiences see the real performer can change how

harshly women's mistakes are judged, although the animated versions were not simply fairer to women, because they may also have made women's successful shots look slightly less impressive relative to men's.

Same Performance, Different Judgments: Gendered Evaluation Across Success and Failure

Evaluations of work are also evaluations of whoever appears to have done it. Would the same success earn credit, and the same mistake draw blame, as readily if a woman appeared to be responsible? Across financial advice, adviser misconduct, and surgical referrals, the answer seems to be asymmetric: Women are penalized more than men after comparable failures or misconduct (Abel et al., 2024, among male participants; Egan, Matvos, & Seru, 2022; Sarsons, 2022), but not before outcomes are known (Abel et al., 2024) or after successes (Abel et al., 2024; Sarsons, 2022). Similarly, leaders in gender-incongruent roles are accorded less status after a mistake than their gender-congruent counterparts (Brescoll, Dawson, & Uhlmann, 2010). The question is old (Swim & Sanna, 1996), but the comparison it calls for, the same failure by a woman and by a man, does not occur in the field. No two real failures are identical, and evaluations of real people entangle what was done with who appears to have done it, and with reputation and context. Experiments can hold the failure fixed, but they typically convey it as information rather than show it (e.g., Abel et al., 2024; Brescoll et al., 2010). We construct the comparison in professional tennis, a setting where men and women do the same job under the same rules of play. We re-render points with motion-capture animation so that the same play can be shown as performed by its original players, by avatars of either gender, or by a robot. We ask whether the avatars remove any extra credit or blame that women's plays receive, or whether that credit or blame follows the gender the avatar displays.

What Theory Leaves Open

Theory does not settle whether a woman's play should be judged more leniently or more harshly than the same play by a man. Where men are expected to perform better, as in sports (Biernat & Vescio, 2002), this stereotype can push evaluations in opposite directions (Biernat & Kobrynowicz, 1997). In assimilation, judgments fall in line with the stereotype, as under double standards, by which a woman must do more than a man to be

credited with ability and a man is given more latitude before incompetence is inferred (Biernat, Fuegen, & Kobrynowicz, 2010; Biernat & Kobrynowicz, 1997; Foschi, 2000). Contrast, in which judgments reverse the stereotyped difference, arises under shifting standards, when each performer is judged against the expectations for their own gender (Biernat & Kobrynowicz, 1997; Biernat, Manis, & Nelson, 1991). A good play can then earn more praise coming from a woman, of whom less is expected (Biernat & Vescio, 2002), and, by the same logic, an error may look worse "for a man" (Biernat & Kobrynowicz, 1997, p. 545). Common-rule scales (e.g., rankings or dollar amounts) tend to reveal assimilation, whereas subjective scales reduce, mask, or sometimes reverse it (Biernat et al., 2010; Biernat & Kobrynowicz, 1997; Biernat & Manis, 1994). Our rating task points both ways: The scale is subjective, but each participant judges men's and women's plays in succession and without having to justify their ratings, conditions that may favor assimilation (Biernat & Manis, 1994; Foschi, 2000). Which effect prevails is therefore open, and the evidence above suggests that the answer may differ between successes and failures.

Both accounts, however, rest on expectations attached to the performer's gender category (Biernat & Kobrynowicz, 1997), and gender effects have been found when gender was conveyed by a name, a photograph, a computer-generated face, or a computer's voice (Abel et al., 2024; Biernat et al., 2010; Biernat & Vescio, 2002; Nass, Moon, & Green, 1997). If the displayed gender is enough, whatever leniency or harshness the original broadcast footage elicits should follow the gender the avatar displays. An avatar of the player's own gender, performing the same play, should then preserve that leniency or harshness, and only removing or equalizing the displayed gender should eliminate it. If the effect requires the original footage instead, whether because the real player conveys gender more strongly than an avatar or for another reason, every avatar should remove it, including one of the player's own gender. We test these implications separately for successes and failures.

Holding the Performance Fixed

Our stimuli are real points from the 2023 Wimbledon Championships. From each of eight matches, four men's and four women's, we took one point ending in a winner (a shot that wins the point outright) and one ending in an error labeled unforced in the official highlights (a mistake not forced by the opponent's shot). Of the four avatar versions, an avatar of the original player's gender (the same-gender avatar) removes the real player from view but keeps the displayed gender. The robot has no gendered appearance. The two fixed-gender avatars, one male and one female, each perform all eight points, so every point displays the same gender (Figure 1). Each participant saw one version, rated either the eight winners or the eight unforced errors, and was not told whether the clips were original or re-rendered. The two implications agree that the robot and the fixed-gender avatars should remove whatever leniency or harshness the original footage elicits. They diverge on the same-gender avatar, which should preserve it if the displayed gender is enough and remove it otherwise. This test assumes that participants perceive the avatar's gender, which we did not measure. Because the men's and women's points are different plays, the gap between them in any one version (the origin gap) confounds play with judgment, so we compare this gap for the same points across versions. Attributing a change in the gap to the original footage assumes that re-rendering's other consequences, such as imperfect motion capture or a general cost of synthetic presentation, affect men's and women's points alike.

The closest study in sports hid the gender cue. Once every person in clips of soccer goals was blurred so that viewers could not tell men from women, men's goals were no longer rated higher than women's (Gomez-Gonzalez, Dietl, Berri, & Nesseler, 2024). But blurring also makes the play harder to judge and hides the real players and their gender together, and that study applied it only to successes. Recoloring players' skin in clips of real soccer tackles has set a demographic cue on a fixed action, but on the real players' bodies (Nesseler, Gasparetto, Naidenova, & Parshakov, 2025). Re-rendering keeps the play

in view and separates the real player's appearance from the displayed gender, and we apply it to failures as well as successes.

Overview of the Present Research

We conducted two preregistered experiments in which U.S. adults recruited on Prolific ($N = 3{,}729$) rated video recordings of professional tennis points. Experiment 1 concerns winners, rated for impressiveness; Experiment 2 concerns unforced errors, rated for severity. The two experiments share the same setting, stimulus pipeline, and design; apart from their participants, they differ only in the clips and the rating item, so the results for successes and failures can be compared. For theories of gendered evaluation, the experiments test, on real plays held fixed, whether judgments of women's successes and failures show assimilation or contrast, and whether the gender a performer displays is enough to elicit either.

Transparency and Openness

Both experiments were preregistered together at the American Economic Association's registry for randomized controlled trials (AEA RCT Registry) on July 12, 2024 (last updated July 25, 2024), before data collection (AEARCTR-0014001, https://www.socialscienceregistry.org/trials/14001; a copy of the preregistration is also on OSF). The preregistered hypotheses were nondirectional and concerned moderation by participant gender and the comparisons of the original footage with the robot, male-avatar, and female-avatar conditions; all other analyses are exploratory. We departed from the preregistered analysis in two respects. First, we omitted the preregistered covariates (participant demographics and characteristics of the points); because every participant rated all eight points, adding them leaves each change in the origin gap unchanged to four decimal places. Second, we tested moderation by participant gender with the four avatar versions pooled rather than in each preregistered comparison; tested separately, it was not significant in any of the six comparisons (ps $\geq .12$).

We report how we determined our sample sizes, all data exclusions (the exclusion

rules were not preregistered), all manipulations, and all measures in the studies, following the Journal Article Reporting Standards (Appelbaum et al., 2018). Data, analysis code, and materials are available on OSF (Pan, Gomez-Gonzalez, & Dietl, 2026). The analysis code, which reproduces Figures 2 and 3 and every number in the text that is computed from the participant data, is also provided as Supplemental Material. The stimulus videos contain copyrighted broadcast footage; during review, they are available through the same link.

Analyses were run in Python 3.14.3 (Python Software Foundation, 2026) with statsmodels 0.14.6 (Seabold & Perktold, 2010) and SciPy 1.17.1 (Virtanen et al., 2020); the mixed models were fitted with code written for this design. Both experiments were approved by the Human Subjects Committee of the Faculty of Business, Economics and Informatics at the University of Zurich (OEC IRB #2024-060). We used Claude (Anthropic, 2026) to help write and check the analysis code, prepare the figures, and edit the manuscript; the authors verified all outputs against the data and take full responsibility for the content.

Experiment 1: Winners

In Experiment 1, participants rated how impressive each winner was, and we ask how the origin gap depends on presentation.

Method

Participants and Design

Participants were recruited on Prolific with quotas for sex, age, and ethnicity matched to the U.S. population and were paid £0.50 for a study advertised as taking about 3 minutes. They took part online, on their own devices and without contact with the experimenters. Experiment 1 was run on August 3–5, 2024, after most of the data for Experiment 2 had been collected. We sampled the general adult public rather than tennis experts because our question concerns how lay viewers evaluate recorded performances. Participants gave informed consent on the first page of the online survey, before any clip

was shown, after reading a consent form that described the study as an evaluation of tennis performance. Of 2,004 completed sessions, 1,997 matched Prolific records, 1,986 were approved submissions, and 1,964 reported watching all eight clips to completion. We further excluded 67 participants who gave the same rating to all eight points; the conclusions are unchanged when they are retained. None of these exclusions was preregistered. The analysis sample is $N = 1{,}897$, with no missing ratings. By Prolific's records of sex, 944 were female and 953 male; their mean age was 43.0 years ($SD = 15.2$). By Prolific's simplified ethnicity categories, 67.1% were White, 11.2% Black, 5.2% Asian, 9.6% mixed, and 7.0% other, and 89.7% were born in the United States (country of birth was missing for 9). More than half (56.4%) held a bachelor's degree or higher, the median household income before taxes was \$60,000–\$79,999, and 29.7% reported never watching tennis. Gender identity, Hispanic or Latino origin, sexual orientation, disability, and immigration history beyond country of birth were not collected. The preregistered power analysis targeted about 400 participants per condition, giving 90% power to detect an interaction of $f^2 = 0.032$ at $\alpha = .01$; recruitment therefore stopped once 2,000 Prolific submissions had been approved. Given the standard errors in the final samples, and with the eight points treated as fixed, each experiment had 80% power (two-sided $\alpha = .05$) to detect a change in the origin gap of 0.15–0.17 rating points ($d = 0.10$–0.11) between the original footage and any one avatar version. For the four avatar versions pooled, the detectable change was 0.12–0.13 rating points ($d = 0.08$).

The survey software (Unipark EFS Survey) randomly assigned participants to one of five presentation conditions ($n = 347$–396 per condition): original footage; a same-gender avatar, in which each point was performed by the avatar of the original player's gender; a robot avatar; a male avatar performing all eight points; or a female avatar performing all eight points. Every participant rated all eight points, four played by men and four by women. The design was thus a 5 (presentation condition, between subjects) × 2 (original player's gender, within subjects) mixed design.

Stimuli

We drew stimuli from the 2023 Wimbledon Championships, quarterfinals onward, on the two main courts. From each of eight matches (four men's, four women's), we selected one point ending in a winner, as recorded in the official point statistics, from official highlight footage with standardized camera angles. We designed a male and a female avatar character (Blender 4.1); a robot character, chosen to have no gendered appearance, completed the set. Each point was re-rendered with Wonder Studio (Wonder Dynamics), an AI-assisted visual-effects tool that performs markerless motion capture on the original video and composites the characters into the broadcast scene, leaving the court, crowd, and camera view unchanged. Ball trajectory, court position, and outcome were thus identical across versions, and the players' movements were reproduced by motion capture, so that, by design, only the performer's appearance differed (Figure 1). All clips were presented without sound.

Procedure and Measures

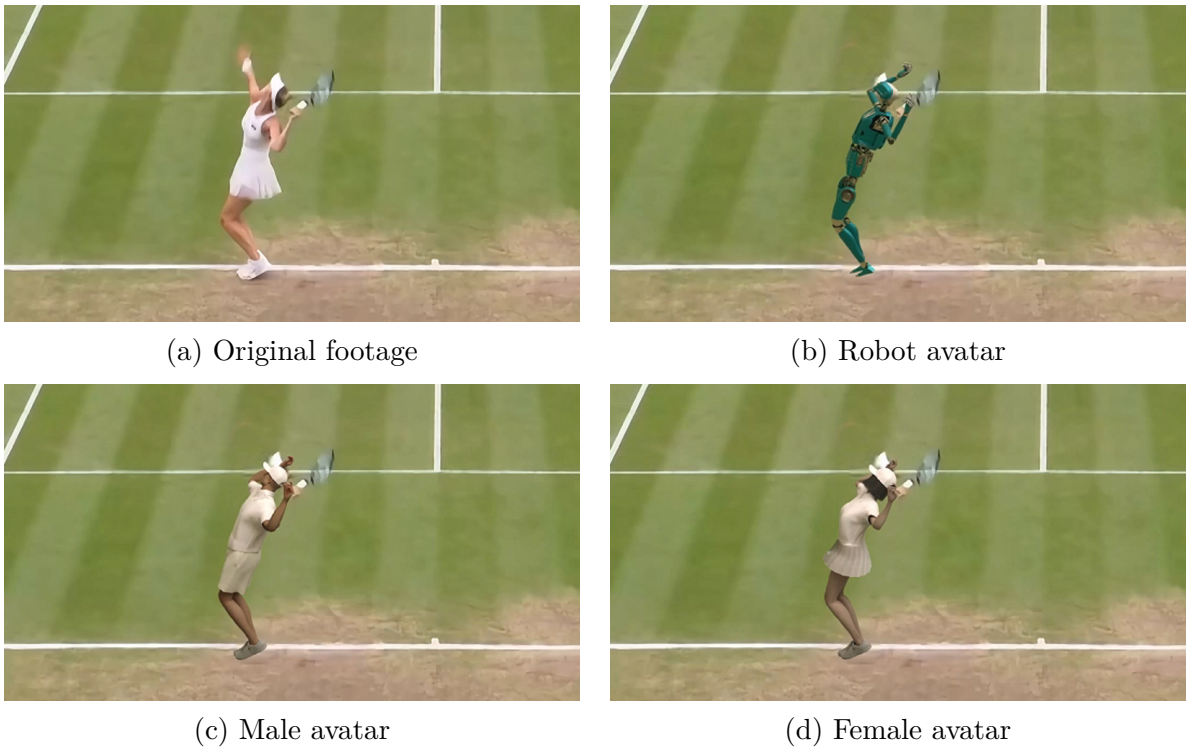
Participants watched the eight points in an independently randomized order and rated each immediately after viewing it. Clips were introduced only as tennis highlights; no condition's clips were described as original or as re-rendered. After each clip, participants answered "How impressive is the win of this point?" on a scale from *Very Unimpressive* to *Very Impressive*, coded 1–7 so that higher values indicate a more impressive win. Age, sex, and ethnicity came from Prolific profiles; education, household income, and how often participants watch tennis (*Never* to *Daily*) were self-reported. The survey ended with these questions; participants were not debriefed about the re-rendering or about the study's focus on gender.

Results

The origin gap is each participant's mean rating of the four women-origin points (those originally played by women, whoever performs them on screen) minus their mean rating of the four men-origin points. Did removing the real players change it? Figure 2

Figure 1

The Same Point With the Performer Re-Rendered (Enlarged Detail)



(a) Original footage (b) Robot avatar

(c) Male avatar (d) Female avatar

Note. Enlarged crops (about 2.4×) of the near player in one stimulus point; the original player is a woman. Participants saw the full broadcast frame, in which the players appear much smaller and the court, officials, and crowd are visible. Re-rendering replaced only the two players and left the rest of the frame unchanged. The same-gender condition reuses the avatars in (c) and (d), matched to each original player's gender. Stills from broadcast footage of the 2023 Wimbledon Championships. During review, the full videos are available on OSF.

displays the mean ratings (panel a) and the gap in the original footage minus the gap in each avatar version (panel b); the comparisons with the robot, male-avatar, and female-avatar conditions were preregistered. We compared conditions by the difference in their mean gaps (or mean ratings), with participant-clustered standard errors and two-sided z tests at $\alpha = .05$, not adjusted for multiple comparisons. Pooled estimates weight the avatar versions by sample size, a Wald test compares the four avatar versions, and d is an estimate divided by the standard deviation of all ratings in the experiment.

Where a conclusion rests on the absence of a difference, we also report an equivalence test (two one-sided tests; Lakens, 2017) against bounds of ±0.10 rating points. These bounds, which were not preregistered, correspond to 0.06–0.07 *SD* of the ratings and

to about a third of the largest origin gap we observed (0.29 rating points, Experiment 2). Like the other participant-level tests, the equivalence tests treat the eight points as fixed, asking whether a result would hold for other participants watching these points. Of the equivalence tests reported below, two passed (one in each experiment); with points as a random factor (Judd, Westfall, & Kenny, 2012), which asks whether a result would also hold for other points, neither did (ps $\geq .13$). These random-factor analyses are linear mixed models with crossed random effects for participants and points and a random slope of presentation for points, estimated by restricted maximum likelihood with Satterthwaite degrees of freedom.

With the original footage, women-origin points were rated less impressive than men-origin points, $b = -0.105$, 95% CI $[-0.18, -0.03]$, $p = .005$, $d = -0.07$. Re-rendering did not shrink this gap. Every avatar version showed a numerically wider gap, and the change in the gap was significant in the two fixed-gender conditions (male avatar: $b = 0.128$, 95% CI $[0.02, 0.23]$, $p = .018$, $d = 0.09$; female avatar: $b = 0.135$, 95% CI $[0.03, 0.24]$, $p = .010$, $d = 0.10$) but not in the same-gender or robot conditions (ps $\geq .117$). The four changes in the gap did not differ significantly from one another, $\chi^2(3) = 3.25$, $p = .355$. The gap with the same-gender avatar lay between the gaps with the original footage and with the other three avatars. It differed significantly from neither (original footage minus same-gender avatar: $b = 0.046$, $p = .414$; same-gender avatar minus the other three avatars: difference $= 0.070$, $p = .143$), but it could not be shown to be equivalent to either (90% CIs $[-0.047, 0.138]$ and $[-0.009, 0.148]$; equivalence tests $p = .168$ and $p = .262$). Pooled across the four avatar versions, the difference was $b = 0.099$, 95% CI $[0.02, 0.18]$, $p = .018$, $d = 0.07$. The gap was therefore smallest in the original footage. Because re-rendering could compress ratings toward the scale midpoint, we also computed a scale-free index, the share of each participant's pairs of one women-origin and one men-origin point in which the women-origin point was rated higher (ties counted as half). It showed a nonsignificant difference in the same direction (46.2% with the original footage

vs. 44.3% with avatars, $p = .063$). With points treated as a random factor, the pooled difference was not significant either ($p = .246$). Re-rendering also lowered rating levels overall (men-origin points: $b = -0.194$, $p < .001$; women-origin points: $b = -0.293$, $p < .001$), consistent with a general cost of synthetic presentation in judged impressiveness. The difference did not vary significantly with participant gender, which we preregistered as a moderator (men minus women participants: $b = -0.130$, $p = .119$), but an equivalence test could not rule out a participant-gender difference of 0.10 rating points or more (90% CI $[-0.267, 0.007]$, equivalence test $p = .641$).

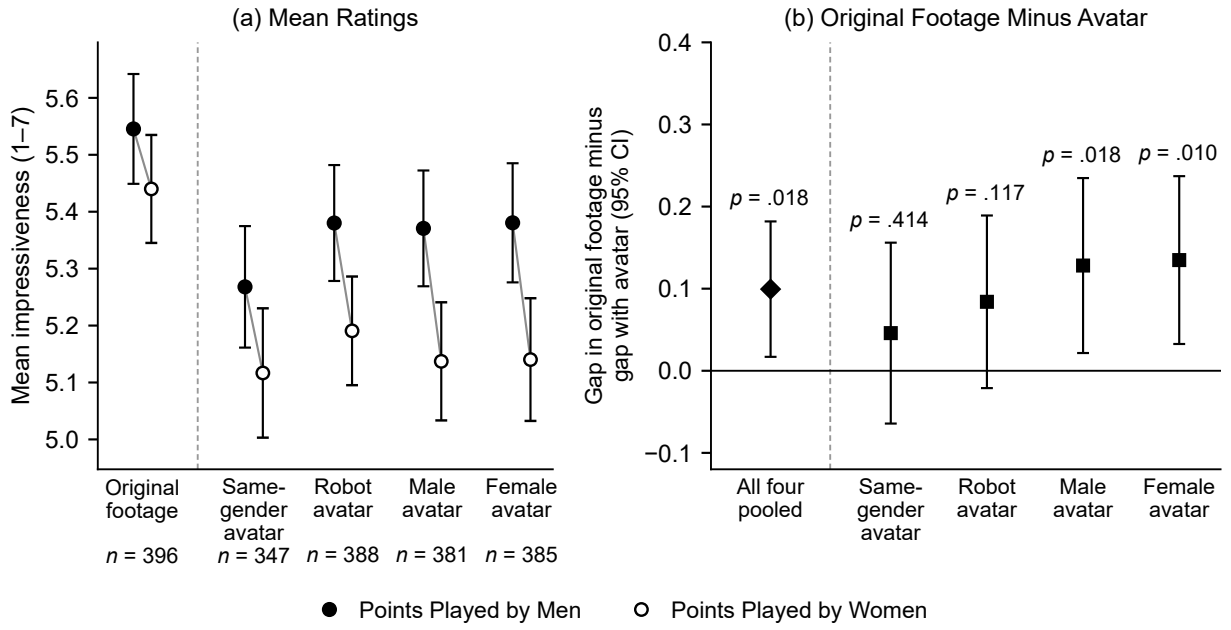
The two fixed-gender conditions showed identical content dressed as a man or as a woman. Ratings did not differ significantly, $b = 0.006$, 95% CI $[-0.13, 0.14]$, $p = .927$ (model with point fixed effects), but they could not be shown to be equivalent (90% CI $[-0.109, 0.122]$, equivalence test $p = .092$). The origin gap differed by less than a tenth of a scale point between the two conditions (difference $= -0.007$, $p = .901$; 90% CI $[-0.095, 0.082]$, equivalence test $p = .041$). On these winners, the displayed gender alone produced no significant change in the ratings and changed the gap by less than a tenth of a scale point.

Discussion

In the original footage, women-origin winners trailed men-origin winners by less than when avatars performed the same points. This change in the gap was modest and significant against only two of the four avatar versions, the fixed-gender avatars, and the four changes did not differ significantly from one another. Swapping the displayed gender changed the gap by less than a tenth of a scale point, whereas keeping each player's gender on an avatar gave an inconclusive result. The same-gender avatar's gap fell between those of the original footage and the other avatars and differed reliably from neither. Any gain for women-origin winners in the original footage, relative to the avatars, was thus small and unreliable.

Figure 2

Ratings of Winners by Presentation Condition, Experiment 1



Note. (a) Mean impressiveness of the four points played by men (filled) and the four played by women (open); the line joining them is the gap. (b) The gap in the original footage minus the gap in each avatar version, and pooled across all four (diamond); positive values mean that the points played by women were rated relatively more impressive in the original footage. Error bars are 95% confidence intervals from participant-clustered standard errors; the intervals and p values treat the eight points as fixed. $N = 1{,}897$; n = number of participants in each condition.

Experiment 2: Unforced Errors

Experiment 2 applies the same design to failures. The same eight matches supplied one unforced error each, and participants rated how severe each error was.

Method

Participants and Design

Recruitment, payment, consent, conditions, and design were identical to those of Experiment 1. Experiment 2 was run first, from July 30 to August 1, 2024, and its last places were filled on August 5–7, after Experiment 1 had begun. Of 2,014 completed sessions, 1,958 passed the same completion screening. Experiment 1 had been closed to Experiment 2 participants, but the places filled on August 5–7 were not closed to Experiment 1 participants; the 26 who took part in both experiments were excluded.

Excluding a further 100 participants who gave the same rating to all eight points leaves $N = 1{,}832$, again with no missing ratings. As in Experiment 1, none of these exclusions was preregistered, and the conclusions are unchanged when the excluded participants are retained. By Prolific's records of sex, 925 were female and 907 male; their mean age was 45.3 years ($SD = 15.8$). By Prolific's simplified ethnicity categories, 64.3% were White, 11.9% Black, 6.0% Asian, 10.5% mixed, and 7.3% other, and 86.8% were born in the United States (country of birth was missing for 6). More than half (60.2%) held a bachelor's degree or higher, the median household income before taxes was \$60,000–\$79,999, and 28.8% reported never watching tennis.

Stimuli and Procedure

Stimuli came from the same eight matches and were re-rendered in the same way. From each match we selected one point presented as an unforced error in the title of its official highlight video. In the official point statistics, six of the eight are recorded as unforced errors and two, one men's and one women's, as forced errors (errors induced by the opponent's shot). The procedure was identical to that of Experiment 1, except that participants answered "How severe is this error?" on a scale from *Very Minor* to *Very Severe*, coded 1–7 so that higher values indicate a more severe error.

Results

Did removing the real players change the origin gap for errors? Figure 3 displays the mean ratings (panel a) and the gap in the original footage minus the gap in each avatar version (panel b). With the original footage, women-origin errors were rated more severe than men-origin errors, $b = 0.292$, 95% CI $[0.21, 0.37]$, $p < .001$, $d = 0.18$. Every avatar version reduced this gap. The changes in the gap ranged from $b = 0.152$ (95% CI $[0.04, 0.27]$, $p = .010$, $d = 0.09$) to $b = 0.204$ (95% CI $[0.09, 0.32]$, $p < .001$, $d = 0.13$) and did not differ significantly from one another, $\chi^2(3) = 1.08$, $p = .782$. The same-gender avatar, which keeps each original player's gender, reduced the gap about as much as the versions that remove or equalize the displayed gender ($b = 0.155$, 95% CI $[0.04, 0.27]$,

$p = .006$, $d = 0.10$). The gap that remained with the same-gender avatar was within a tenth of a scale point of the gap with the other three avatars (difference = 0.022, $p = .624$; 90% CI $[-0.052, 0.097]$, equivalence test $p = .044$). Pooled across the four avatar versions, the difference was $b = 0.172$, 95% CI $[0.08, 0.26]$, $p < .001$, $d = 0.11$; the gap that remained was 0.121, 95% CI $[0.08, 0.16]$, $p < .001$, about 40% of the original-footage gap. The extra severity attached to women-origin errors was thus largest in the original footage and dropped by more than half under every avatar version. Looking at men-origin and women-origin errors separately locates the change. Re-rendering did not significantly alter the rated severity of men-origin errors ($b = -0.081$, $p = .245$) but lowered that of women-origin errors ($b = -0.253$, $p < .001$). On the scale-free index, the women-origin error was rated more severe in 56.2% of each participant's pairs with the original footage and in 52.6% with avatars ($p < .001$). Because 50% means no difference, re-rendering removed 58% of the excess over 50%. With points treated as a random factor, however, the pooled difference was not significant ($p = .137$). As in Experiment 1, the difference did not vary significantly with participant gender ($b = -0.045$, $p = .629$), but an equivalence test could not rule out a participant-gender difference of 0.10 rating points or more (90% CI $[-0.200, 0.109]$, equivalence test $p = .281$).

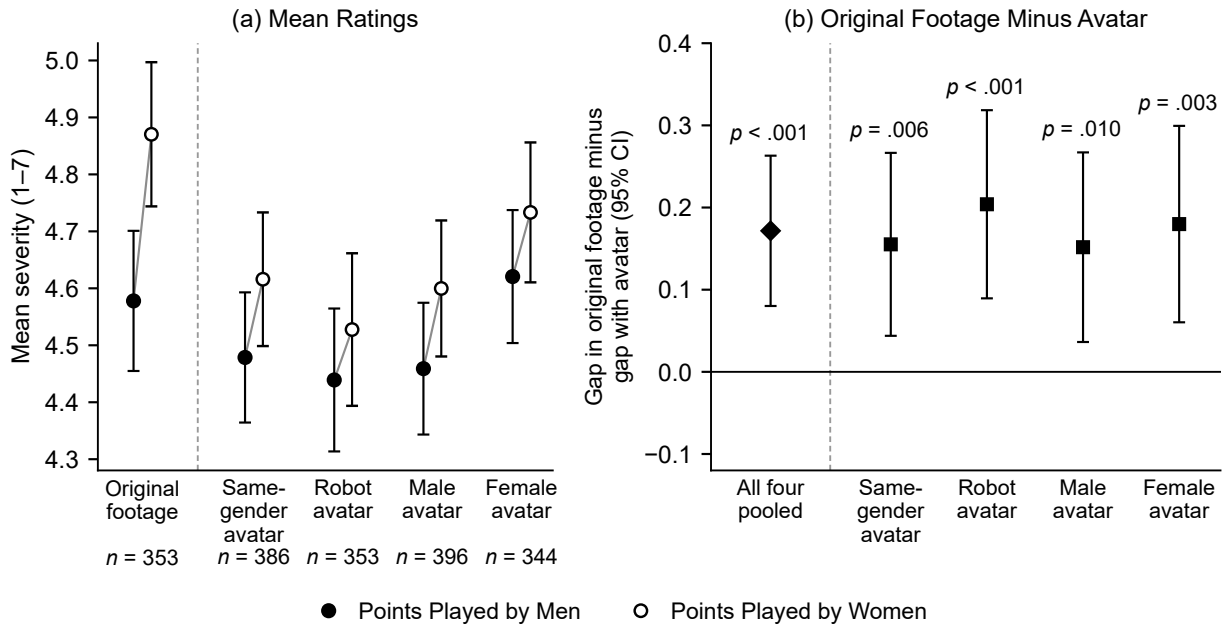
Dressing identical content as a woman rather than a man produced a small, nonsignificant increase in rated severity, $b = 0.148$, 95% CI $[-0.01, 0.30]$, $p = .065$ (model with point fixed effects). The ratings with the two fixed-gender avatars could not be shown to be equivalent either (equivalence test $p = .724$), so a small effect of the displayed gender alone on judged severity cannot be ruled out. The origin gap did not differ significantly between the two fixed-gender conditions (difference = -0.028, $p = .643$), but it could not be shown to be equivalent (90% CI $[-0.128, 0.072]$, equivalence test $p = .118$).

Across the two experiments, the pooled difference was not significantly larger for errors than for winners (0.172 vs. 0.099; difference = 0.072, 95% CI $[-0.05, 0.20]$, $p = .251$), but the two could not be shown to be equivalent (90% CI $[-0.031, 0.176]$, equivalence test

$p = .329$). We also estimated a single effect common to both experiments, by which the original footage raised women-origin plays relative to men-origin plays on either scale (winners more impressive, errors more severe). With points as a random factor, this common effect was $b = 0.127$, $p = .055$. Pooled across experiments, the difference did not vary significantly with whether participants watched any tennis (watchers minus nonwatchers: $b = 0.024$, $p = .725$).

Figure 3

Ratings of Unforced Errors by Presentation Condition, Experiment 2



Note. (a) Mean severity of the four errors made by men (filled) and the four made by women (open); the line joining them is the gap. (b) The gap in the original footage minus the gap in each avatar version, and pooled across all four (diamond); positive values mean that the errors made by women were rated relatively more severe in the original footage. Error bars are 95% confidence intervals from participant-clustered standard errors; the intervals and p values treat the eight points as fixed. $N = 1{,}832$; n = number of participants in each condition.

Discussion

In the original footage, women-origin errors were rated more severe than men-origin errors by about three tenths of a scale point, and more than half of that extra severity disappeared whenever an avatar performed the errors. Removing the real players while keeping each player's gender (the same-gender avatar) reduced the gap about as much as

the other versions did, and swapping the displayed gender did not significantly change it. The part of the extra severity that the avatars removed was thus tied to the original footage of the real players, not to the plays themselves or to the displayed gender of an avatar. Unlike the pattern for winners in Experiment 1, this reduction was significant against every avatar version.

General Discussion

In two preregistered experiments, we held each play constant and varied who appeared to perform it. In the original footage, women's unforced errors were rated more severe than men's, and every avatar version cut this extra severity by more than half, including avatars of the players' own gender. Winners showed a weaker pattern in the opposite direction. Women's winners trailed men's by less in the original footage than with the avatars, but the change was significant against only two of the four avatar versions, the fixed-gender avatars. The original footage thus worked against women when they erred but not when they succeeded. The evidence is clearer for failures than for successes, but because the two effects did not differ significantly in size, it does not show that the original footage matters more for one than the other.

Theoretical Implications

For failures, the extra severity that the avatars removed is consistent with assimilation. Under double standards, incompetence is inferred more readily from a woman's failure than from a man's (Foschi, 2000), and leaders in gender-incongruent roles are accorded less status after a mistake than gender-congruent leaders (Brescoll et al., 2010). For successes, the small and unreliable gain for women's winners in the original footage, relative to the avatars, points toward contrast, as shifting standards predict (Biernat & Kobrynowicz, 1997). Together, the two accounts can yield this combination, for instance if a woman's success, unlike her failure, is seen as exceptional (Biernat & Kobrynowicz, 1997; Foschi, 2000). Alternatively, the original footage may have made women's plays, relative to men's, look more extreme in both directions. This would follow

if women's ability were seen as more variable than men's, an account that Sarsons (2022) argued against for surgeons because physicians did not respond differently to good outcomes from male and female surgeons. Because the two effects did not differ significantly in size, our data cannot separate these readings, and all of them assume that ratings of a play carry inferences about the player, which we did not measure.

For errors, where the effect was clearer, the results also bear on what elicits it. Provided that participants perceived the gender of the small avatar figures, which we did not measure, displaying the player's gender on our avatars was not enough. An avatar of the player's own gender reduced the extra severity about as much as the robot and the fixed-gender avatars did; the gap that remained with it was within a tenth of a scale point of the gap with those three combined. The result does not mean that participants ignored the avatars' gender. The same errors were rated slightly, though not significantly, more severe when the female avatar performed them than when the male avatar did. What the avatars did not reproduce, with the players' movements reproduced by motion capture, is how the real players were judged. One possibility is that the original footage conveys gender more strongly, for instance by inviting the subcategorization that women in domains with a male default are thought to attract, that is, being categorized as a "female player" rather than simply a "player" (Schulte, Weitzel, Alves, & Lammers, 2026). That account, however, would also cost women's winners credit, which the original footage did not. Another possibility is that synthetic performers fall short of the human likeness on which mind ascription (believing that a figure has a humanlike mind) depends, at least for faces (Deska, Almaraz, & Hugenberg, 2017). If that dependence is categorical for whole performers, as it is for faces, this route would also fit the finding that the robot and the other three avatar versions did not differ significantly in how much they reduced the extra severity.

For research on gender bias, the results suggest that how a performer is presented matters, not only which gender is displayed. Experiments usually convey gender by a label,

a name, or a picture on an otherwise fixed record. AI-generated presenters can now also deliver the same talk as a woman or as a man. Franzen and Wöhner (2026), whose AI presenters showed no bias against women, suspected that the presenters' artificiality concealed a gender effect, although the authors' real presenters showed no such bias either. Our results suggest that a synthetic performer can indeed remove an effect present in the original footage. For errors, varying only the displayed gender would have found a small, nonsignificant effect and missed the one in the original footage. Presentation choices can change judgments of other recorded events as well. Videotaped confessions, for instance, seem more voluntary when the camera focuses on the suspect rather than on the interrogator (Ratcliff, Lassiter, Schmidt, & Snyder, 2006; Ware, Lassiter, Patterson, & Ransom, 2008). Where evaluations concern recorded performances, re-rendering offers a complementary design that varies the performer while holding the performance fixed.

Practical Implications

For practice, these results suggest that showing avatars instead of the real players does not simply work in women's favor. In our experiments, the avatars removed more than half of the extra severity attached to women's errors, but they also lowered the rated impressiveness of winners overall, women's if anything more than men's. Re-rendering is also not a form of blinding. A screen at an audition hides the musician while the jury still hears the playing (Goldin & Rouse, 2000). Re-rendering instead replaced the real players with animated figures that reproduced their movements, so participants saw the same play in a synthetic presentation, and our results do not show what hiding the performer alone would do. Where synthetic versions of real performances are shown, for instance in sports media, their net effect on how women's successes and failures are judged would need to be tested in that setting. Finally, because women's and men's points were different plays, these results do not show whether women's or men's play is better and should not be cited as evidence on that question.

Limitations and Future Directions

The avatars removed the real players together with everything else that only the original footage shows, including the players' differing skin tones (the male and female avatars each had a single skin tone), so we cannot say which property of the footage matters. We did not measure whether participants recognized the players. Recognition seems unlikely to be required, because the players appeared as small figures in the full broadcast frame, the clips had no sound, and the effect did not differ significantly between participants who watch tennis and those who never do. We also did not measure whether participants perceived the avatars' gender in such small figures. No clip was described as original or re-rendered, so the effect did not require telling participants that a clip was real, although they may have inferred it. What people are told about the source of the same work can itself change how they evaluate it (Jago, Carroll, & Lin, 2022; Shank, Stefanik, Stuhlsatz, Kacirek, & Belfi, 2023). A label experiment that varies only what participants are told about the same footage would separate knowing that a performer is real from seeing the original footage. Every avatar also copied the original player's movements, which may themselves carry gender. The robot and the fixed-gender avatars therefore equalized gendered appearance rather than every gender cue, and the gap that remained with them may reflect such cues as well as differences between the plays.

Our comparison also assumes that re-rendering, apart from removing the real players, affected men's and women's points alike, which we could not test. Differences between the male and female avatar designs cannot explain the change in the gap, because the gap also changed in the fixed-gender conditions, where one avatar performed all eight points. Differences in how well the points were rendered could, however. In both experiments, re-rendering lowered the ratings of women's points more than men's, which is what a rendering that made women's points look less striking would do, for instance if their movements were captured less faithfully. Independent ratings of how faithfully each point was rendered would test this. Finally, successes and failures were different points,

rated on different items by different samples rather than compared within participants.

Constraints on Generality

Several constraints bound these conclusions (Simons, Shoda, & Lindsay, 2017). Each experiment used eight points. When points were treated as a random factor (Judd et al., 2012), the difference between the original footage and the avatars was not significant in either experiment or in both combined, and neither equivalence result held. Our conclusions therefore concern these points, and generalizing them requires more points, players, and tournaments. We intend the results to apply to lay U.S. adults viewing recorded tennis. The participants were U.S. adults on Prolific, recruited with quotas for sex, age, and ethnicity matched to the U.S. population. They were more educated than U.S. adults in general, and whether the results hold in other populations is untested. Lay participants may differ from experts, who might also recognize the players, although expertise did not reduce gendered bias in judges' legal decisions (Miller, 2019). Each participant gave single-item, unincentivized ratings, and zero-sum choices, such as team selection, can diverge from such subjective ratings (Biernat & Vescio, 2002). The data were collected in 2024, and growing familiarity with synthetic video, or changing attitudes toward women's sports, could change the results. Finally, professional tennis is a domain with a male default. Whether the original footage matters as much where women are expected to excel remains open.

Replications should keep four features of the design: The same play appears in every version, the clips are silent, participants are not told whether a clip is original or re-rendered, and the original footage shows the full broadcast frame, in which the players are small figures. Adding sound, telling participants that a clip is real, or showing the players at closer range may change the results. Neither participant gender nor tennis viewing significantly moderated the effect; for participant gender, the equivalence tests could not rule out moderation as large as the effect itself. The changes in the gap did not differ significantly between the four avatar versions, but each gender was represented by a

single avatar, so other avatar designs are untested.

Conclusion

In the original footage, women's errors drew extra severity, and more than half of it disappeared when avatars performed the same errors, even avatars of the players' own gender. On successes, the original footage did not work against women; if anything, women's winners trailed men's by less than with avatars. In these experiments, an avatar's displayed gender, if participants perceived it, did not reproduce how the real players' errors were judged. In settings that use synthetic versions of real people, such as sports media, whether to show the real performer is a design choice. Our results suggest that this choice is not neutral, because it can change how much extra severity women's errors draw.

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